TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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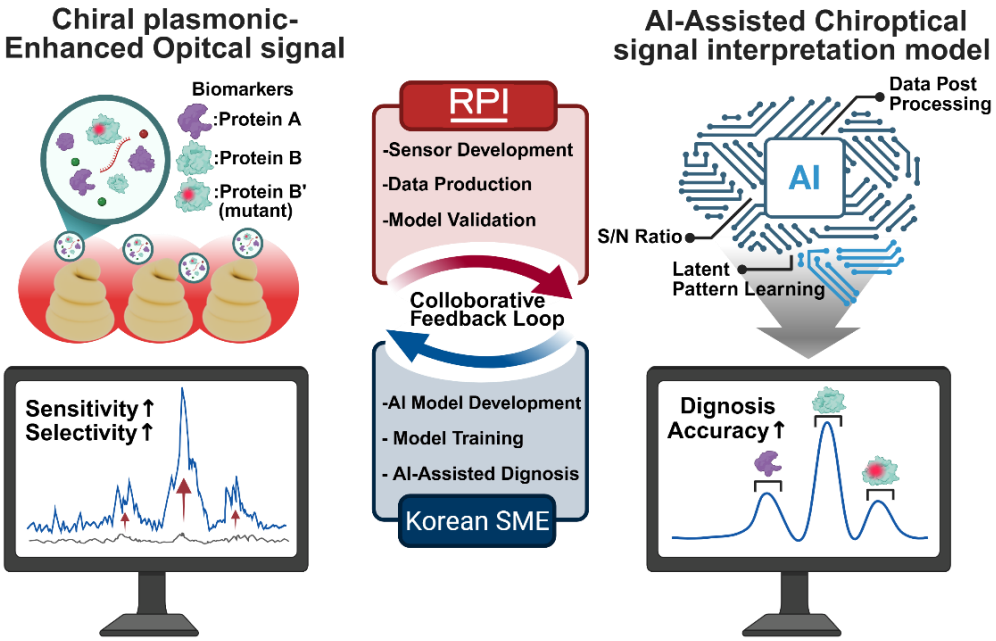
2. Technology Overview

Title of Proposed Technology	AI-Driven Decision Framework for High-Throughput Chiroptical Fingerprint Extraction and Bioanalysis			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio·medical		*Chemicals*	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	*Bio-signal Measurement / Diagnostic Devices*	*Fine Chemicals*	*Nano-Application Technologies*
Concept of Proposed Technology	Chirality is a fundamental physical property that governs biomolecular structure and function, yet robust and comparable chiroptical fingerprints have not been systematically established for many clinically relevant biomarkers.[1–3] If reliably extracted and interpreted, chirality-resolved fingerprints could provide a new diagnostic dimension by capturing subtle structural and conformational alterations associated with disease states, therapeutic response, or pathological progression.[1,4–6] This project advances an AI-driven decision-making framework that transforms complex, high-dimensional chiroptical signals derived from chiral plasmonic platforms into quantitative, interpretable fingerprint representations.[7,8] By enabling robust classification and decision-level interpretation of structure-sensitive optical signatures, the proposed framework establishes a scalable analytical foundation for chiral bioanalysis and diagnostic inference.[9,10]			
Rationale for Joint R&D Project	The systematic interpretation of chiroptical fingerprints poses intrinsic challenges due to the high dimensionality, nonlinearity, and experimental sensitivity of structure-sensitive optical signals.[11–13] Historically, chiroptical measurements have often remained at the level of observing condition-specific signal changes, without standardized fingerprint definitions or reproducible decision frameworks. Overcoming these limitations requires not only advanced experimental capabilities to generate high-quality, information-rich chiroptical data, but also sophisticated AI-driven methodologies capable of extracting, organizing, and interpreting complex signal patterns in a robust and scalable manner.[7,8,10] This project therefore necessitates a tight and iterative integration of experimental chiral plasmonic fingerprint generation and AI-driven decision-level interpretation, which cannot be effectively			

	achieved within a single organization. In the proposed collaboration, the research institute specializes in the development and operation of high-fidelity experimental platforms for generating reliable and structure-sensitive chiroptical fingerprints under controlled and biologically relevant conditions.[11,14–19] In parallel, the Korean SME leads the development, optimization, and deployment of AI-driven decision models for high-throughput fingerprint extraction, classification, and interpretation. Through this complementary role structure, experimental validation continuously informs AI model refinement, while AI-driven insights guide subsequent experimental design and parameter selection. This closed-loop experiment–AI synergy enables systematic convergence toward reproducible, decision-ready chiroptical fingerprints, enhancing both analytical robustness and practical scalability. As a result, the joint R&D framework establishes an efficient pathway for translating complex chiroptical signals into standardized, interpretable fingerprint representations suitable for downstream bioanalytical and diagnostic applications.
Key Words	1. Chiral Plasmonic Biosensing 2. Chiroptical Fingerprinting 3. AI-Driven Signal Interpretation 4. Multi-Biomarker Analysis 5. Experimental–AI Closed-Loop Framework

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (2-3)	**End of the Project**	Level (4-5)
R&D Objectives	***Challenges*** Although chiral plasmonic structures provide strong signal amplification, quantitative interpretation remains challenging. Key challenges include: • **Multidimensional signal composition**, involving spectra, polarization states, and intensity asymmetries • **Nonlinear signal behavior** that cannot be reduced to single parameters or fixed thresholds • **Signal crosstalk** arising from complex biological sample compositions In practical industrial and clinical environments, **biological and experimental variability act simultaneously**. This variability includes differences in sample preparation, measurement conditions, patient-to-patient heterogeneity, and disease or physiological state. As a result, identifying the true origin of optical signal changes becomes difficult. Moreover, sensitivity to structural chirality increases interdependencies and nonlinearities among signal components. Consequently, conventional rule-based, threshold-based, or empirical analysis methods show limited robustness and reproducibility. **AI-based analytical approaches are therefore essential** to extract meaningful features from multidimensional data and to translate chiral plasmonic sensitivity into scalable bioanalytical technologies. ***Research goals*** The goal of this research is to develop an **AI-assisted platform for accurate and consistent interpretation of chiroptical signals** (including CD, SERS, etc.). The specific objectives are to: • Learn multidimensional optical signals arising from chiral plasmonic–biomarker interactions • Automatically extract latent patterns and features reflecting structural and chiral variations • Establish a quantitative link between optical signal changes and underlying biological states • Develop an AI framework capable of integrative analysis of diverse chiral optical datasets Ultimately, the platform will move beyond analysis of known biomarkers. It will enable **identification of new signal-based indicators and discovery of potential novel biomarkers** associated with disease states or biological changes. Through this capability, the proposed research aims to establish a new paradigm for AI-based interpretation of chiral plasmonic optical signals.			

R&D Target Specifications	The target specifications of this R&D project are defined to quantitatively validate the performance and applicability of the proposed AI-assisted chiral plasmonic signal analysis platform. • Establishment of a **disease- and biomarker-specific chiroptical signature library**, derived from chiral plasmonic amplification of multiple representative biomarkers associated with selected diseases (e.g., cancer or Alzheimer's disease). • Development of an **AI-assisted interpretation model capable of classifying predefined biomarker signatures with an accuracy exceeding 85%** under controlled laboratory conditions. • Achievement of **at least a 30% reduction in signal interpretation variability** compared to conventional rule-based or empirical analysis methods, through AI-assisted feature extraction and analysis. • Demonstration of the platform's capability to process **multidimensional chiroptical datasets**, including circular dichroism spectra and polarization-dependent optical signals, within a unified analysis framework. • Validation of **robust analytical performance under complex biological conditions**, including heterogeneous samples containing multiple coexisting biomarkers. • Demonstration of **simultaneous recognition and interpretation of multiple biomarkers** using the integrated chiral plasmonic–AI platform. • Identification of **candidate chiroptical indicators or potential novel biomarker signatures** that distinguish patient samples from healthy control samples in proof-of-concept studies. • Validation of the integrated analysis platform at the **laboratory scale (TRL 4)**, with a clearly defined pathway for future expansion to clinical or in-line bioanalytical applications.
R&D Plan	This research will be carried out in a stepwise manner to interpret chiroptical signals amplified on chiral plasmonic platforms using AI-assisted approaches, and to extend these capabilities toward disease diagnosis and discovery of new biomarker candidates. The overall research framework and role-sharing structure between RPI and the Korean SME are illustrated in Figure.  **Figure** Synergistic collaboration between experimental chiral plasmonic biosensing and AI-assisted signal interpretation through an iterative feedback loop. **Phase 1: Acquisition of Disease-Specific Chiroptical Signatures and Construction of an AI-Based Library** In Phase 1, the project will focus on selected diseases (e.g., cancer or Alzheimer's disease) and systematically acquire chiroptical signals amplified on chiral plasmonic structures from multiple disease-related biomarkers. AI-assisted analysis methods will be applied to these optical signals to identify biomarker-specific chiroptical signatures (e.g. CD, SERS, etc.), which will be organized into a structured data library.

	Key activities in this phase include: • Acquisition of chiroptical signals from diverse disease-related biomarkers on chiral plasmonic platforms • Signal preprocessing procedures, including normalization, baseline correction, and noise characterization • AI-based extraction of characteristic signal patterns and definition of biomarker-specific signatures • Systematic organization and accumulation of defined chiroptical signatures into a library **As an outcome of this phase, a disease- and biomarker-specific chiroptical signature library, along with an initial AI-based interpretation model, will be established.** The resulting library and AI model will serve as a scalable analytical foundation that can be extended to additional diseases in subsequent studies. **Phase 2: Sensor and AI Model Advancement through Application to Complex Biological Environments** In Phase 2, the chiral plasmonic-based biosensor and AI analysis models developed in Phase 1 will be applied to real patient samples or systems that mimic complex biological environments. Based on these results, both the sensor platform and the AI models will be further refined and advanced. In this phase, the roles of the sensor and the AI model are addressed separately but in a complementary manner. From the sensor perspective, the focus will be on: • Reliable acquisition of chiroptical signals from real patient samples or complex biological environments • Verification of reproducible signal measurement under conditions involving multiple coexisting biomarkers From the AI analysis perspective, the following objectives will be pursued: • Improvement of AI model performance to account for signal overlap (crosstalk) and variability in complex sample environments • Development of analysis capabilities for simultaneous recognition, separation, and interpretation of multiple biomarkers • Comparative analysis between patient samples and healthy control samples to identify disease-associated signal features Through these efforts, the project aims to go beyond analysis of predefined biomarkers and to **explore new chiroptical indicators or potential novel biomarker candidates that distinguish patient groups from healthy controls**. This phase will demonstrate the practical applicability of the chiral plasmonic–AI biosensing platform and its potential for operation in complex biological environments.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will serve as the core experimental partner in this collaboration, responsible for establishing the chiral plasmonic experimental platform and providing experimental validation to support the development and refinement of AI models led by the Korean SME partner. The specific roles and responsibilities of the research institute include: • Design and implementation of chiral plasmonic–based biosensing platforms and establishment of experimental protocols • Acquisition of chiroptical signals from diverse disease-related biomarkers (e.g., cancer,

	Alzheimer's disease) using chiral plasmonic structures • Generation of multidimensional chiroptical datasets, including circular dichroism (CD) and polarization-dependent optical signals • Experimental characterization and physical interpretation of biomarker-specific chiroptical signatures • Experimental evaluation of AI models developed by the Korean SME, including assessment of classification accuracy, reproducibility, and robustness • Provision of experiment-based feedback to support iterative improvement and refinement of AI models • Design and execution of experiments using complex biological environments and patient-derived samples to validate AI model applicability Through these activities, the research institute will ensure the experimental validity of the AI models and support their reliable operation in realistic bioanalytical settings.
Expected R&R of Korean SME Partner	The Korean SME partner will take primary responsibility for the development of AI-based chiroptical signal interpretation models, leveraging experimental datasets provided by the research institute to build, optimize, and validate advanced analytical algorithms. The specific roles and responsibilities of the Korean SME partner include: • Design and development of AI models for interpretation of multidimensional chiroptical datasets generated by the research institute • Construction of biomarker-specific signature classification models using machine learning and deep learning approaches • Improvement of signal separation and interpretation performance under complex biological conditions involving multiple coexisting biomarkers • Enhancement of AI model generalization to account for biological and experimental variability • Implementation of AI-based classification and prediction models for comparative analysis of patient samples and healthy control samples • Data-driven exploration of new chiroptical indicators and potential novel biomarker candidates Through this collaboration, the Korean SME partner will establish AI analysis technologies specialized for chiral plasmonic biosensing data and lay the groundwork for future expansion into bioanalytical and diagnostic applications.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes** Through this joint R&D project, an AI-assisted analysis platform capable of interpreting chiral plasmonic optical signals will be established, overcoming the limitations of conventional empirical signal analysis approaches. The expected technical outcomes include: • Construction of a **biomarker-specific chiroptical signature library** derived from chiral plasmonic amplification of disease-related biomarkers • Development of an **AI-based analysis platform** for robust interpretation of complex, multidimensional chiroptical signals • Establishment of **multi-biomarker recognition and analysis capabilities** under conditions where multiple biomarkers coexist • Experimental validation of AI model performance using **real patient samples and complex biological environments** • Identification of **new chiroptical indicators and potential novel biomarker candidates** that have not been previously defined

These outcomes will demonstrate the scientific and technical feasibility of AI-assisted chiral plasmonic signal analysis and provide a scalable foundation for extension to additional diseases and bioanalytical applications.

Expected Impact

The outcomes of this joint R&D project are expected to generate broader technological, industrial, and economic impacts beyond the immediate research scope.

1) Technological and Industrial Impact

- Advancement from conventional concentration-based biosensing toward **structure- and chirality-based bioanalytical technologies**
- Improved reliability and reproducibility of complex optical signal interpretation through **AI-assisted analytical frameworks**
- Establishment of a **differentiated convergence technology portfolio** combining chiral plasmonic sensing and AI-based signal analysis

2) Commercialization and Application Potential

- The developed AI analysis models can be extended to **biosensor data analysis software, analytical modules, or integrated platforms**
- The project provides a pathway for **Korean SMEs to develop AI-based bioanalytical solutions** with practical application potential
- The collaboration establishes a foundation for **technology transfer, follow-up projects, and commercialization strategies**

3) Industrial and Economic Impact

- Creation of **high value-added analytical technologies** applicable to bio-diagnostics, precision medicine, and biomanufacturing quality control
- Strengthening the technological competitiveness of **Korean SMEs in AI–bio convergence fields** and supporting future global market entry
- Contribution to the growth of the **domestic bio and healthcare innovation ecosystem** through data-driven bioanalysis technologies

6. References

1. Hwang, M., Jung, H. & Kim, J.-Y. Chirality Quantification for High-Performance Nanophotonic Biosensors. *Small Methods* **n/a**, 2500112 (2025).

2. Liu, Y., Wu, Z., Armstrong, D. W., Wolosker, H. & Zheng, Y. Detection and analysis of chiral molecules as disease biomarkers. *Nat. Rev. Chem.* **7**, 355–373 (2023).

3. John, N. & Mariamma, A. T. Recent developments in the chiroptical properties of chiral plasmonic gold nanostructures: bioanalytical applications. *Microchimica Acta* **188**, 424 (2021).

4. Hao, C., Xu, C. & Kuang, H. Chiral probes for biosensing. *Chemical Communications* **59**, 12959–12971 (2023).

5. Kabashin, A. V, Kravets, V. G. & Grigorenko, A. N. Label-free optical biosensing: going beyond the limits. *Chem. Soc. Rev.* **52**, 6554–6585 (2023).

6. Ma, W., Xu, L., Wang, L., Xu, C. & Kuang, H. Chirality-Based Biosensors. *Adv. Funct. Mater.* **29**

1805512 (2019).

7. Coca-Lopez, N. *et al.* Artificial Intelligence-Powered Raman Spectroscopy through Open Science and FAIR Principles. *ACS Nano* **19**, 38189–38218 (2025).

8. Kothari, R. *et al.* Raman spectroscopy and artificial intelligence to predict the Bayesian probability of breast cancer. *Sci. Rep.* **11**, 6482 (2021).

9. Takita, H. *et al.* A systematic review and meta-analysis of diagnostic performance comparison between generative AI and physicians. *NPJ Digit. Med.* **8**, 175 (2025).

10. D'Adderio, L. & Bates, D. W. Transforming diagnosis through artificial intelligence. *NPJ Digit. Med.* **8**, 54 (2025).

11. Choi, W. J. *et al.* Chiral phonons in microcrystals and nanofibrils of biomolecules. *Nat. Photonics* **16**, 366–373 (2022).

12. Im, S., Mousavi, S., Chen, Y.-S. & Zhao, Y. Perspectives of chiral nanophotonics: from mechanisms to biomedical applications. *npj Nanophotonics* **1**, 46 (2024).

13. Gumus, E., Bingol, H. & Zor, E. Nanomaterials-enriched sensors for detection of chiral pharmaceuticals. *J. Pharm. Biomed. Anal.* **221**, 115031 (2022).

14. Xu, L. *et al.* Enantiomer-dependent immunological response to chiral nanoparticles. *Nature* **601**, 366–373 (2022).

15. Kang, Y.-T. *et al.* Chiroptical detection and mutation analysis of cancer-associated extracellular vesicles using microfluidics with oriented chiral nanoparticles. *Matter* **7**, 4373–4389 (2024).

16. Kumar, P. *et al.* Photonically active bowtie nanoassemblies with chirality continuum. *Nature* **615**, 418–424 (2023).

17. Kim, J. Y. *et al.* Direct-write 3D printing of plasmonic nanohelicoids by circularly polarized light. *Proc. Natl. Acad. Sci. U. S. A.* **121**, (2024).

18. Cha, M. *et al.* Unifying structural descriptors for biological and bioinspired nanoscale complexes. *Nat. Comput. Sci.* **2**, 243–252 (2022).

19. Ohnoutek, L. *et al.* Third-harmonic Mie scattering from semiconductor nanohelices. *Nat. Photonics* **16**, 126–133 (2022).

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization